Results outline

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1 Wildchrokie

1.1 Growing season length and conditions

1. Across the three years and all species, the growing season length varied from 90 days to 173 days.
2. The year with longest growing season length on average was 2019 with 141.49 days, and the shortest was 2020 with 135.87 days.
3. The hottest spring period of the three years (MAM) was 2020 with a mean mean temperature of 9.05°C, and the coldest, 2019 with a mean mean temperature of 8.72°C.
4. The hottest summer period of the three years (JJA) was 2020 with a mean temperature of 23.07°C, and the coldest, 2019 with a mean temperature of 22.38°C.
5. The driest spring period of the three years (MAM) was 2019 with a PDSI value of 2.74, and the wettest, 2020 with a PDSI value of 1.32.
6. The driest summer period (PDSI) of the three years (JJA) was 2019 with a PDSI value of 4.02, and the wettest, 2020 with a PDSI value of -2.35.
7. The spring period with the highest precipitation of the three years (MAM) was 2018 with a precipitation of 120.87, and the lowest, 2020 with a precipitation of 91.77.
8. The summer period with the highest precipitation of the three years (JJA) was 2019 with a precipitation of 115.33, and the lowest, 2020 with a precipitation of 66.37.

1.2 Phenology observations

1. Leafout dates across all years and species varied from 112 to 157
2. On average, the earliest leafout date was in 2019 for all species (A. incana: 120.86, B alleghaniensis:123, B. papyrifera: 123.33 and B. populifolia: 121. The latest leafout date for all species was in 2020 (A. incana: 148.68, B alleghaniensis:141.32, B. papyrifera: 136.57 and B. populifolia: 133.83).
3. On average, the earliest budset date was in 2019 for A. incana(A. incana: 263.29 and B alleghaniensis: 258), but in 2018 for B. papyrifera: 266.83 and B. populifolia: 266.29. The latest budset date for all species was in 2020 (A. incana: 280.88, B alleghaniensis:264.05, B. papyrifera: 279.43 and B. populifolia: 282).

1.3 Ring widths

1. The ring width of our species vary from ∞mm to $-\infty$mm.

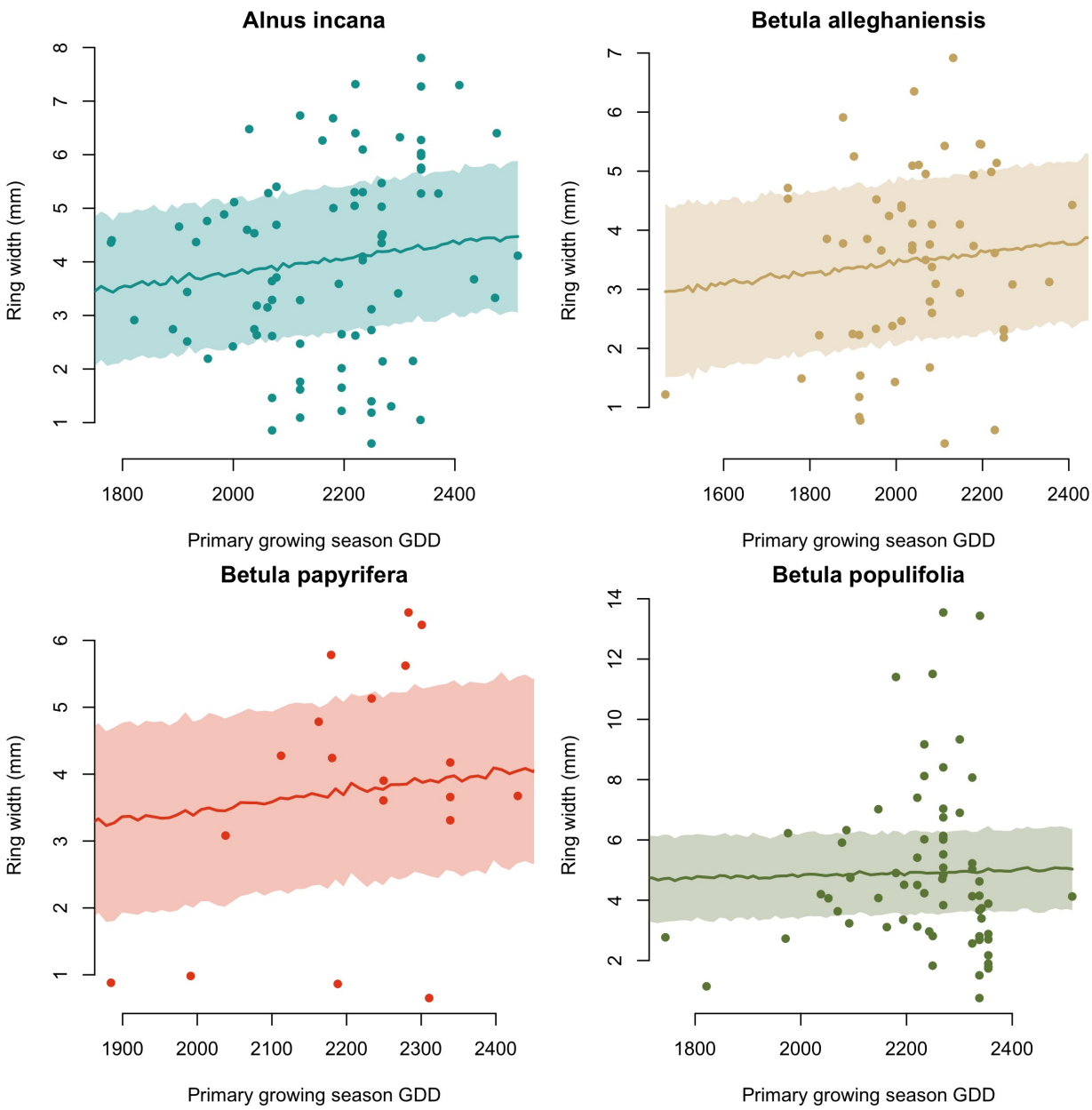

Figure 1: Ring width responses to growing degree days for the primary growing season. The shaded areas represent the 50% confidence intervals. The dots represent the empirical data

1.4 Model results

Our results show similar trends in response to the thermal (GDD) versus the calendar growing season (GSL), with a slightly stronger response to longer GSL compared to increased GDD.

1. A. incana ($\beta = 0.001385$, $UI_{90}[-4.05 \times 10^{-4}, 0.003215]$), B. alleghaniensis ($\beta = 8.75 \times 10^{-4}$, UI_{90}[-0.00123, 0.00293]) and B. papyrifera ($\beta = 0.00142$, UI_{90}[-0.001365, 0.00423]) show a weak response, but seem to grow more with more GDD, while B. populifolia ($\beta = 4.35 \times 10^{-4}$, UI_{90}[-0.00175, 0.002615]) did not respond to increased GDD (Figure 2).

2. A. incana ($\beta = 0.0374$, $UI_{90}[0.0132, 0.0615]$) is the only species for which its 90% intervals don't overlap with zero suggesting that it grows more with longer growing season. B. alleghaniensis ($\beta =$

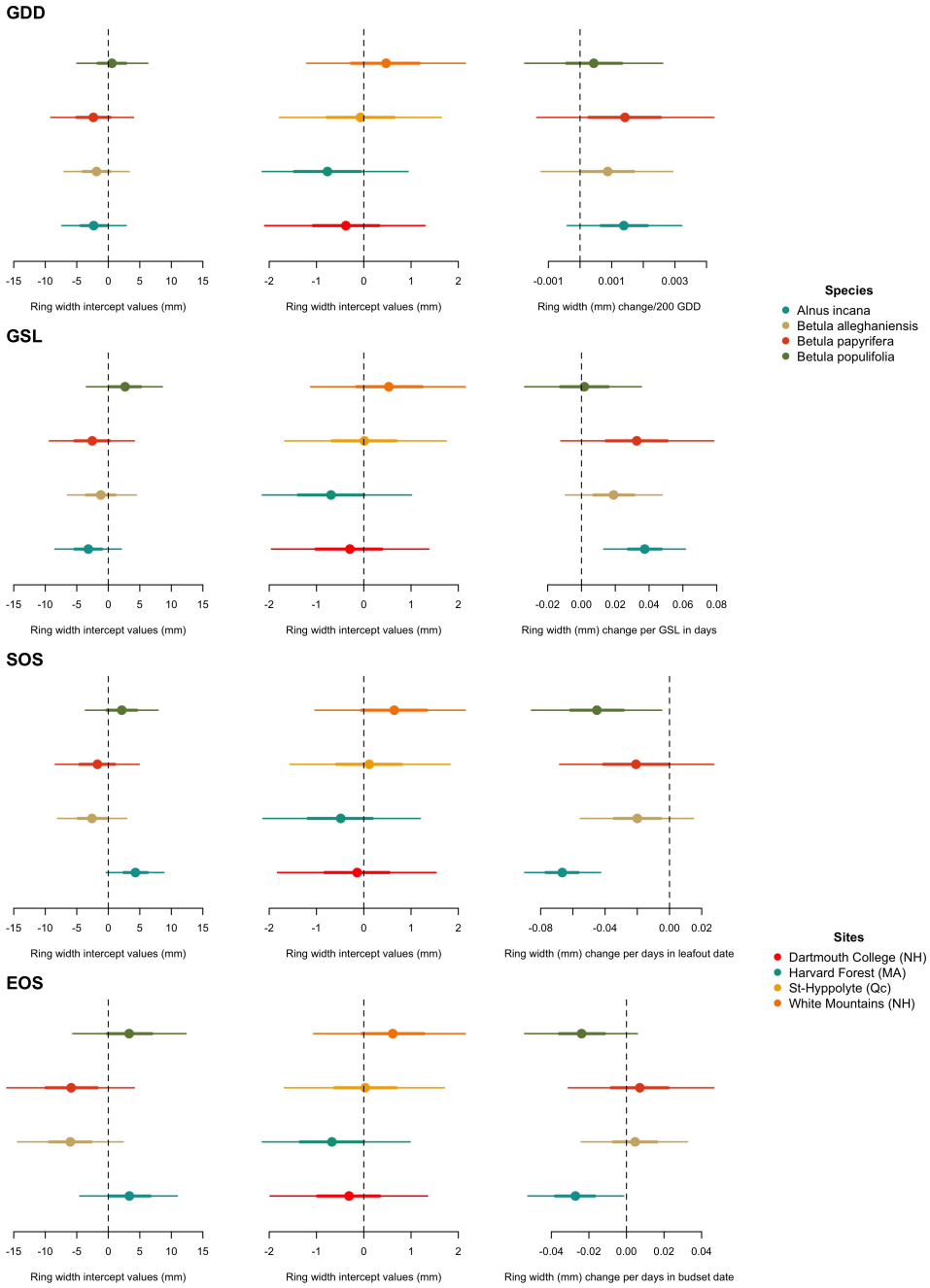

Figure 2: Model parameters across the different predictors

0.019, UI_{90}[-0.0095, 0.0477]) and B. papyrifera (β = 0.0327, UI_{90}[-0.0122, 0.0783]) show a weaker response, but seem to grow more in response to longer seasons, while B. populifolia (β = 0.0017, UI_{90}[-0.0336, 0.0354]) did not respond to the GSL (Figure 2).

3. Ring width decreased with a later SOS for A. incana (β = -0.0666, UI_{90}[-0.0899, -0.0428]) and B. populifolia (β = -0.0451, UI_{90}[-0.0857, -0.0049]). Whereas B. alleghaniensis (β = -0.02, UI_{90}[-0.0554, 0.0149]) and B. papyrifera (β = -0.0208, UI_{90}[-0.0682, 0.0275]) show a weaker response, but seem to grow less with later SOS. (Figure 2).

4. Ring width decreased with a later EOS for A. incana (β = -0.0272, UI_{90}[-0.0525, -0.0015]). B. populifolia showed a weaker response but appears to also grow less with a later EOS (β = -0.0239, UI_{90}[-0.0542, 0.0058]). B. alleghaniensis (β = 0.0046, UI_{90}[-0.0241, 0.0325]) and B. papyrifera (β = 0.0071, UI_{90}[-0.031, 0.0465]) didn't respond to EOS. (Figure 2).

5. For the provenance with the GDD model, we detected a weak growth difference between G. College (α_{site} = -0.379, UI_{90}[-2.097, 1.299]), S. Hypolyte (α_{site} = -0.07, UI_{90}[-1.784, 1.635]) or W. Mountains (α_{site} = 0.471, UI_{90}[-1.208, 2.187]). Only HF grew less (α_{site} = = -0.768, UI_{90}[-2.439, 0.936]) (Figure 2)

coringTreespotters

Supplemental results

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